SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY COURSE
CODE: CSA5116

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COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage : 20%

*QUESTIONS:* *Total Marks: 50*

**Annexure A**

## Scenario-Based

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***Code of Conduct:***

*I Sarika K S 192521039 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*



***Signature of the student:***

*Annexure A*

*Scenario-based*

### 1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

**A.** Explain why this issue occurs.
**B.** Suggest a more secure block cipher mode and justify your choice based on the scenario.

### 2. *Choosing an Encryption Algorithm for Banking System*

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

**A.** Identify the most suitable algorithm for this scenario.
**B.** Justify your choice based on security strength and efficiency.

### 3. *Key Exchange in an Unsecured Network*

Two users want to establish a secure communication channel over a public network using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

**A.** Explain how the key exchange works in this scenario.
**B.** Identify a possible security threat and suggest a solution to mitigate it.

### 4. *Public Key Cryptography for Email Security*

An organization plans to implement RSA for secure email communication.

**A.** Describe how RSA can be used for encryption and decryption in this scenario.
**B.** Explain how key distribution and management play a role in maintaining security.

### 5. *Cryptography for IoT Devices*

A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

**A.** Analyze the constraints of IoT devices.
**B.** Recommend a suitable cryptographic approach and justify your answer.

## QUESTION 1:

## SECURE DATA TRANSMISSION USING BLOCK CIPHER

Scenario: A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

### A. Explain why this issue occurs

### 1. Identification

- ✓ The main issue in the given scenario is the leakage of information caused by using Electronic Codebook (ECB) mode for encrypting financial data. ECB is a block cipher mode of operation in which each plaintext block is encrypted independently using the same secret key.
- ✓ When a company transmits repeated financial information, such as account numbers, transaction amounts, or customer details, the same plaintext blocks may occur multiple times. ECB produces identical ciphertext blocks whenever the plaintext blocks are identical and the same key is used. As a result, patterns in the original data can remain visible in the encrypted data.
- ✓ Therefore, ECB mode does not provide sufficient protection against the analysis of repeated data patterns.

### 2. Explanation

1. A block cipher is a cryptographic algorithm that encrypts data in fixed-size blocks. For example, AES processes data in blocks of 128 bits. A block cipher mode of operation determines how these blocks are encrypted when the complete message contains multiple blocks.
2. ECB is one of the simplest modes of operation. It divides the plaintext into separate blocks and encrypts every block independently. The same secret key is used for all blocks.
3. The encryption process can be represented as:
4. $$\mathbf{C_i = E_k(P_i)}$$
5. Where $P_i$ represents the i-th plaintext block, $C_i$ represents the corresponding ciphertext block, and $E_k$ represents the encryption algorithm using secret key K.
6. Suppose a company encrypts financial data containing repeated values. If two plaintext blocks are identical, their ciphertext blocks will also be identical.
7. For example, if Transaction A appears twice, both occurrences produce the same ciphertext block under ECB when encrypted with the same key. An attacker who observes the encrypted message can identify repeated patterns, even without knowing the encryption key.

## This creates several security problems:

- Pattern leakage: ECB reveals repeated plaintext blocks through repeated ciphertext blocks. This can expose information about the structure of the original message.
- Information analysis: An attacker may compare ciphertexts from multiple messages and identify repeated sections. If the attacker already knows part of the original data, the comparison may reveal additional information.
- Lack of block dependency: Each block is encrypted independently. The encryption of one block does not depend on the previous block, so ECB does not hide repeated patterns across blocks.
- Unsuitable for sensitive financial data: Financial records contain structured and sometimes repeated information. Exposing patterns can help attackers analyze the data and potentially infer details about transactions or message formats.
- No integrity protection: ECB encryption alone does not detect whether an attacker has modified, removed, or rearranged ciphertext blocks.

For these reasons, ECB is not recommended for protecting long or structured messages containing sensitive information.

## 3. Mathematical Formula

$$C_i = E_k(P_i)$$

For two identical plaintext blocks:

$$P_1 = P_2$$

Using the same secret key:

$$E_k(P_1) = E_k(P_2)$$

Therefore:

$$C_1 = C_2$$

This demonstrates that identical plaintext blocks generate identical ciphertext blocks in ECB mode.

## 4. Example

- ✓ Consider a financial application that encrypts transaction records. Two records contain the same repeated data block, such as a standard transaction code. If the same key is used, ECB encrypts both blocks independently and produces identical ciphertext blocks. Although the attacker cannot directly read the transaction code, the repeated ciphertext pattern reveals that the same block appears more than once.
- ✓ If the attacker knows the structure of the financial records, the repeated patterns may help them infer information about the message. Thus, ECB provides confidentiality at the individual-block level but fails to hide repeated patterns in the overall message.

## B. Suggest a more secure block cipher mode and justify your choice based on the scenario

### 1. Identification

- ✓ Cipher Block Chaining (CBC) mode is a better choice than ECB for the company's financial data because it reduces the visibility of repeated plaintext patterns.
- ✓ CBC is a block cipher mode in which each plaintext block is combined with the previous ciphertext block before encryption. For the first block, an Initialization Vector (IV) is used instead of a previous ciphertext block.
- ✓ The IV introduces variation into the encryption of the first block. When the IV is selected correctly, identical plaintext blocks are much less likely to produce identical ciphertext blocks at the same positions in separate messages.
- ✓ For the scenario given, CBC is a suitable improvement over ECB, although modern systems should generally prefer authenticated encryption, such as AES-GCM.

### 2. Explanation

CBC improves upon ECB by creating a dependency between consecutive blocks. In CBC mode, the first plaintext block is combined with an IV using the XOR operation. The result is then encrypted with the secret key.

For every subsequent block, the plaintext is combined with the previous ciphertext block before encryption.

Step 1: Divide the plaintext. The company's financial data is divided into fixed-size blocks $P_1$, $P_2$, $P_3$, …, $P_n$.

Step 2: Select an Initialization Vector. An IV is generated for the message. For CBC, the IV must be unpredictable and should not be reused carelessly. It does not need to be secret.

Step 3: Encrypt the first block. The first plaintext block is XORed with the IV and then encrypted using the secret key.

$$C_1 = E_k(P_1 \oplus IV)$$

Step 4: Encrypt the remaining blocks. Each new plaintext block is XORed with the previous ciphertext block before encryption.

$$C_i = E_k(P_i \oplus C_{i-1})$$

Step 5: Transmit the ciphertext. The encrypted blocks are transmitted to the receiver. The receiver uses the same secret key and the IV to decrypt the message.

The decryption formulas are:

$$P_1 = D_k(C_1) \oplus IV$$

$$P_i = D_k(C_i) \oplus C_{i-1}$$

Here, $D_k$ represents decryption using the secret key.

## 3. Advantages of CBC Mode

- Reduces pattern leakage: CBC combines each plaintext block with the previous ciphertext block, so ciphertext depends on the current plaintext and the previous ciphertext.
- Provides better message confidentiality than ECB: CBC makes repeated plaintext blocks less directly visible when the IV is handled correctly.
- Suitable for multi-block messages: CBC links blocks together instead of encrypting every block independently.
- Uses the same block cipher: CBC can be used with block ciphers such as AES, allowing the company to retain a standard block cipher while changing the mode.
- Supports sequential processing: Blocks can be encrypted in sequence because each block depends on the previous ciphertext block.

CBC also has limitations. It requires padding for messages that are not an exact multiple of the block size, and encryption is sequential. CBC alone does not provide message integrity or authenticity. A separate message authentication code (MAC) and careful implementation are required to detect tampering.

## 4. Comparison Between ECB and CBC

| Feature | ECB Mode | CBC Mode |
|---|---|---|
| Full form | Electronic Codebook | Cipher Block Chaining |
| Encryption method | Blocks encrypted independently | Each block depends on previous ciphertext |
| Initialization Vector | Not required | Required |
| Repeated plaintext blocks | Produce identical ciphertext blocks | Depends on IV and preceding ciphertext |
| Pattern leakage | High for repeated blocks | Reduced when correctly implemented |
| Integrity protection | Not provided | Not provided |
| Suitability for financial records | Not recommended | Better than ECB, but needs integrity protection |

## 5. Mathematical Formula

ECB encryption:

$$C_i = E_k(P_i)$$

CBC encryption:

$$C_1 = E_k(P_1 \oplus IV)$$

$$C_i = E_k(P_i \oplus C_{i-1})$$

The symbol $\oplus$ represents XOR, which combines the bits of two binary values. In CBC mode, even when two plaintext blocks are identical, their ciphertext blocks can differ because the values combined with the plaintext blocks may differ.

## 6. Conclusion

- ECB mode is unsuitable for the company's financial data because it encrypts each block independently and reveals repeated ciphertext patterns. This can help an attacker analyze the structure of the encrypted message and infer information about repeated data.
- CBC mode is a better alternative because it combines each plaintext block with the previous ciphertext block, reducing the visibility of repeated patterns. It uses an Initialization Vector to introduce variation into the first block.
- Therefore, the company should avoid ECB mode and adopt a properly implemented secure encryption scheme. CBC with a correctly generated IV and separate integrity protection is better than ECB. For a new financial transmission system, authenticated encryption such as AES-GCM is generally preferable because it provides both confidentiality and integrity.

## QUESTION 5:

## CRYPTOGRAPHY FOR IoT DEVICES

Scenario: A smart home system uses low-power IoT devices that need secure communication. The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

## A. Analyze the constraints of IoT devices

## 1. Identification

- The main issue in the given scenario is that IoT devices have limited hardware resources but still need strong cryptographic security. IoT devices include smart locks, motion sensors, temperature sensors, smart lights, security cameras, and other connected household equipment.
- Many of these devices operate with small processors, limited memory, low battery capacity, and restricted network bandwidth. The designer must select a cryptographic algorithm that provides strong security without placing excessive demands on the device's resources.
- The two available options are RSA and Elliptic Curve Cryptography (ECC). Both are public-key cryptographic approaches, but their key sizes and computational requirements differ.

## 2. Explanation

- ✓ Internet of Things (IoT) devices communicate over networks to exchange sensor readings, control commands, alerts, and other information. Since these devices may transmit sensitive information or control physical equipment, secure communication is important.
- ✓ For example, a smart door lock must ensure that an unauthorized person cannot read or modify control messages. A temperature sensor should also protect information sent to the smart home controller.
- ✓ However, IoT devices cannot always use the same computing resources as desktop computers or servers. Their limitations must be considered before choosing a cryptographic algorithm.

## 3. Major Constraints of IoT Devices

### a. Limited processing power

Many IoT devices use microcontrollers with relatively low processing capacity. Complex cryptographic calculations can take more time and consume additional processing resources. RSA operations can be expensive, especially private-key operations involving large keys.

ECC can provide comparable security with smaller keys and can be efficient when implemented on suitable hardware.

### b. Limited memory

IoT devices often have restricted RAM and storage. Cryptographic algorithms require memory for keys, intermediate calculations, program code, and temporary data. RSA typically requires larger keys than ECC for comparable security. This can increase storage requirements and the amount of data handled during communication. ECC's smaller keys make it attractive for devices with limited memory.

### c. Limited battery capacity

Some IoT devices operate on batteries for months or years without frequent charging. Heavy cryptographic computation can consume energy and reduce battery life. The designer must consider the energy required for key establishment, encryption, decryption, and authentication. ECC can reduce the amount of data transmitted and can offer efficient cryptographic operations, depending on the implementation and hardware.

### d. Limited communication bandwidth

IoT devices may communicate through low-bandwidth wireless networks. Transmitting large public keys or digital signatures increases communication overhead. RSA generally uses larger keys and signatures than commonly used ECC schemes at comparable security levels. ECC can reduce the size of cryptographic data exchanged between devices.

### e. Need for strong security

IoT devices may control door locks, alarms, cameras, and other important household functions. Attackers could attempt to intercept messages, impersonate devices, or modify commands. The selected cryptographic system must help protect confidentiality, authentication, and integrity. A secure algorithm alone is not sufficient; secure key storage, device authentication, and proper protocol design are also required.

### f. Long-term operation

Many IoT devices remain installed for years. The cryptographic system should support secure key management, updates, and replacement of compromised credentials. The design should avoid weak keys and outdated algorithms. Devices should also be capable of receiving security updates where possible.

### g. Cost and hardware limitations

IoT devices are often designed to be small and inexpensive. They may not include dedicated cryptographic hardware. A lightweight cryptographic implementation can reduce hardware requirements, but it must still be tested for security and performance.

## 4. Impact of These Constraints on Algorithm Selection

- ✓ The constraints show that the designer cannot select an algorithm based only on its theoretical security. The algorithm must also operate efficiently within the available resources. RSA is a well-established public-key cryptosystem, but its relatively large key sizes can increase storage, transmission, and computation costs.
- ✓ ECC provides comparable security with much smaller keys. This makes it attractive for IoT devices that have limited memory, bandwidth, processing power, and battery capacity. Therefore, the constraints of IoT devices support choosing ECC, provided

that the implementation and supported cryptographic protocols meet the system's security requirements.

## B. Recommend a suitable cryptographic approach and justify your answer

## 1. Identification

- ✓ Elliptic Curve Cryptography (ECC) is the recommended choice for the smart home IoT system because it provides strong security with relatively small key sizes and can operate efficiently on resource-constrained devices.
- ✓ ECC is a public-key cryptographic approach based on mathematical operations involving points on elliptic curves over finite fields. Compared with RSA, ECC can achieve comparable security with much smaller keys. This reduces key storage and communication overhead and can improve the suitability of public-key cryptography for small IoT devices.
- ✓ ECC supports several cryptographic functions, including key establishment and digital signatures. For secure communication, the designer can use Elliptic Curve Diffie–Hellman (ECDH) to establish a shared secret and then use a symmetric encryption algorithm such as AES-GCM to protect messages.

## 2. How ECC Can Be Used in the Smart Home System

Consider a smart door lock communicating with a smart home controller.

Step 1: Generate keys. The door lock generates an ECC private key and calculates its corresponding public key. The controller also generates its own key pair.

Step 2: Exchange public keys. The devices exchange their public keys using an appropriate secure protocol. The public keys must be authenticated so that an attacker cannot substitute their own keys.

Step 3: Establish a shared secret. The devices use ECDH to calculate a shared secret from their own private keys and the other device's public key.

Step 4: Derive a symmetric encryption key. A key derivation function converts the shared secret into a suitable symmetric encryption key.

Step 5: Encrypt the communication. The devices use AES-GCM to encrypt and authenticate control commands and sensor data.

Step 6: Verify incoming messages. The receiving device checks the authentication tag before accepting the message. This helps detect unauthorized modifications.

Step 7: Protect the private keys. The private keys are securely stored and never transmitted over the network. Devices should also support appropriate key rotation and security updates.

This approach uses ECC for efficient key establishment and symmetric encryption for protecting the actual data.

## 3. Advantages of ECC for IoT Devices

- Smaller key size: ECC provides comparable security with significantly smaller keys than RSA. For example, a 256-bit ECC key provides approximately the same security level as a 3072-bit RSA key for commonly used, appropriately selected schemes. Smaller keys require less storage and reduce the size of public-key data transmitted between devices.
- Lower communication overhead: Smaller public keys and signatures can reduce the amount of cryptographic data transmitted, which is useful for sensors and battery-powered devices that send frequent messages.
- Efficient resource usage: ECC can offer efficient operations on suitable hardware. Smaller key sizes can also reduce memory requirements and the amount of data processed during key exchange. Actual performance depends on implementation, curve, and processor.
- Reduced energy requirements in suitable implementations: Efficient ECC implementations can reduce energy consumption compared with some RSA-based approaches at equivalent security levels. Energy consumption should be measured on the target hardware.
- Strong security: ECC can provide strong security when suitable curves, key sizes, implementations, and protocols are selected.
- Suitable for resource-constrained hardware: ECC's compact keys make it a practical choice for small connected devices.
- Supports key establishment and authentication: ECC-based protocols can help devices establish shared keys and verify identities through digital signatures.

## 4. Comparison Between RSA and ECC

| Criteria | RSA | ECC |
|---|---|---|
| Full form | Rivest–Shamir–Adleman | Elliptic Curve Cryptography |
| Security basis | Integer factorization | Elliptic-curve discrete logarithm problem |
| Key size | Relatively large | Relatively small |
| Memory requirements | Generally higher at comparable security | Generally lower at comparable security |
| Communication overhead | Larger keys/signatures in common configurations | Smaller keys/signatures in common configurations |
| Computational efficiency | Private-key operations can be resource-intensive | Often efficient on constrained devices |
| Energy usage | May be higher depending on hardware and operation | Can be lower with suitable implementations |
| IoT suitability | Possible, but less attractive for highly constrained devices | Generally more suitable for constrained devices |
| Key establishment | RSA-based key transport is possible in supported protocols | ECDH can establish a shared secret |

| | | |
|---|---|---|
| Authentication | RSA digital signatures | ECC-based digital signatures |

The comparison shows that ECC is generally the more practical choice for a low-power smart home system. Nevertheless, both algorithms can be secure when implemented correctly, and the final selection should consider the actual hardware and protocol requirements.

## 5. Mathematical Formula

A commonly used elliptic-curve equation over a finite field is:

$$y^2 \equiv x^3 + ax + b \pmod{p}$$

Where x and y represent point coordinates, a and b are curve parameters, and p is a prime number defining the finite field. The curve parameters must be selected so that the curve is suitable for cryptographic use.

ECC public-key generation can be represented as:

$$Q = dG$$

Where Q is the public key, d is the private key, G is the publicly defined base point, and dG represents elliptic-curve point multiplication.

For ECDH, suppose Device A has private key dA and public key QA = dA G. Device B has private key dB and public key QB = dB G.

Device A calculates:

$$S_a = d_a Q_\beta$$

Device B calculates:

$$S_\beta = d_\beta Q_a$$

Substituting the public-key equations gives:

$$S_a = d_a(d_\beta G) = d_a d_\beta G$$

$$S_\beta = d_\beta(d_a G) = d_\beta d_a G$$

Therefore:

$$S_a = S_\beta$$

Both devices derive the same shared point without sending their private keys. A key derivation function is then used to produce a symmetric key for secure communication.

## 6. Security Considerations

- Use standardized, secure elliptic curves and suitable key sizes.
- Generate private keys using a secure random-number generator.
- Authenticate exchanged public keys to prevent man-in-the-middle attacks.
- Use a key derivation function after ECDH rather than using the raw shared secret directly.
- Protect private keys from unauthorized access.
- Use authenticated encryption, such as AES-GCM, to protect confidentiality and message integrity.
- Implement secure updates and key rotation to maintain long-term security.

These measures ensure that the use of ECC provides meaningful protection within the overall smart home security system.

## 7. Conclusion

- IoT devices have limited processing power, memory, battery capacity, and communication bandwidth. Therefore, they require a cryptographic approach that provides strong security while minimizing resource usage.
- ECC is the recommended choice over RSA for the given smart home scenario because it offers comparable security with smaller keys, lower communication overhead, and potentially better efficiency on constrained hardware.
- The designer can use ECDH to establish a shared secret, a key derivation function to generate a symmetric key, and AES-GCM to encrypt and authenticate the communication. Proper key management and public-key authentication are also necessary.
- Therefore, ECC combined with authenticated symmetric encryption is a suitable approach for protecting communication between low-power smart home IoT devices.

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**RUBRICS FOR CALCULATION:**

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |